Big Data HomeWork 1

Lekhana Sandra

Step-by-Step Guide for Setting Up AWS Academy Learner Lab

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1. Open AWS Academy Learner Lab from Modules

- Navigate to your **AWS Academy course dashboard**.
- Click on **"Modules"** in the navigation panel.
- Select **"AWS Academy Learner Lab"** and open it.

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2. Start AWS Academy Learner Lab from Modules

- Click **"Start Lab"** to initiate the environment.
- Wait for the lab to **provision resources** and become active.

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3. Choose Two Instances and Assign Names

- Open the **AWS Management Console**.
- Navigate to **EC2 Dashboard**.
- Click **"Launch Instance"** and select **two instances**.
- Assign meaningful names like **"Instance1"** and **"Instance2"** under the **Tags** section.

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4. Generate a Key Pair, Download the .PEM File, Copy the Directory Path, and Open in Terminal

- In the **Key Pair** section, click **"Create new key pair"**.
- Choose **RSA** as the key type and enter a name (e.g., `MyKeyPair`).
- Click **"Create"** and **download the .PEM file**.

Open **Terminal** and navigate to the directory where the `.PEM` file is stored:
cd /path/to/your/keypair

- Set the correct file permissions:
  chmod 400 MyKeyPair.pem

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5. Launch the Instance Successfully

- Click **"Launch Instance"** and wait for it to **initialize**.
- Verify that the instances are in the **"Running"** state in the EC2 Dashboard.

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6. Click on Connect

- In the **EC2 Dashboard**, select an instance.
- Click the **"Connect"** button at the top.
- Choose **"SSH Client"** and follow the connection instructions.

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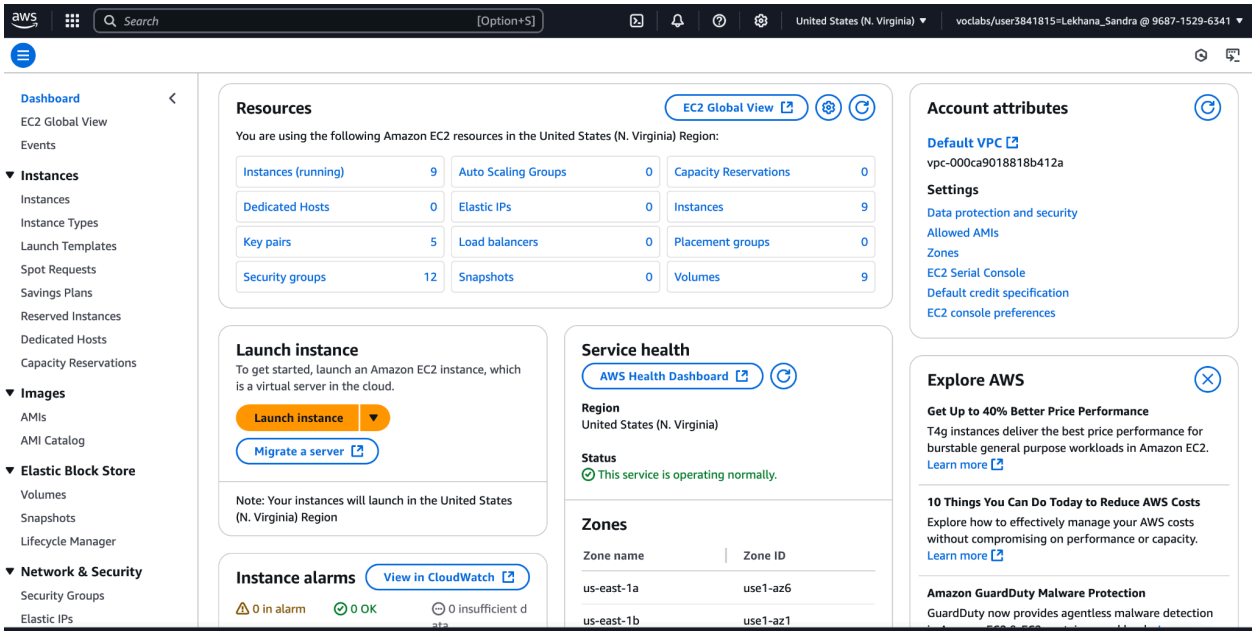
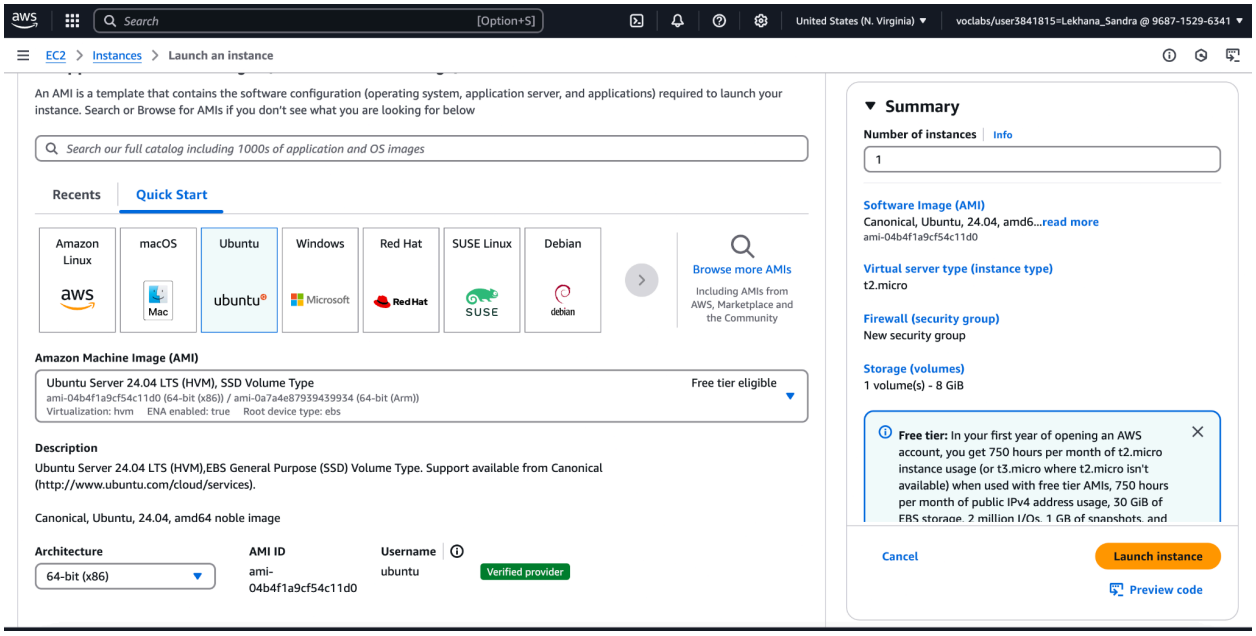
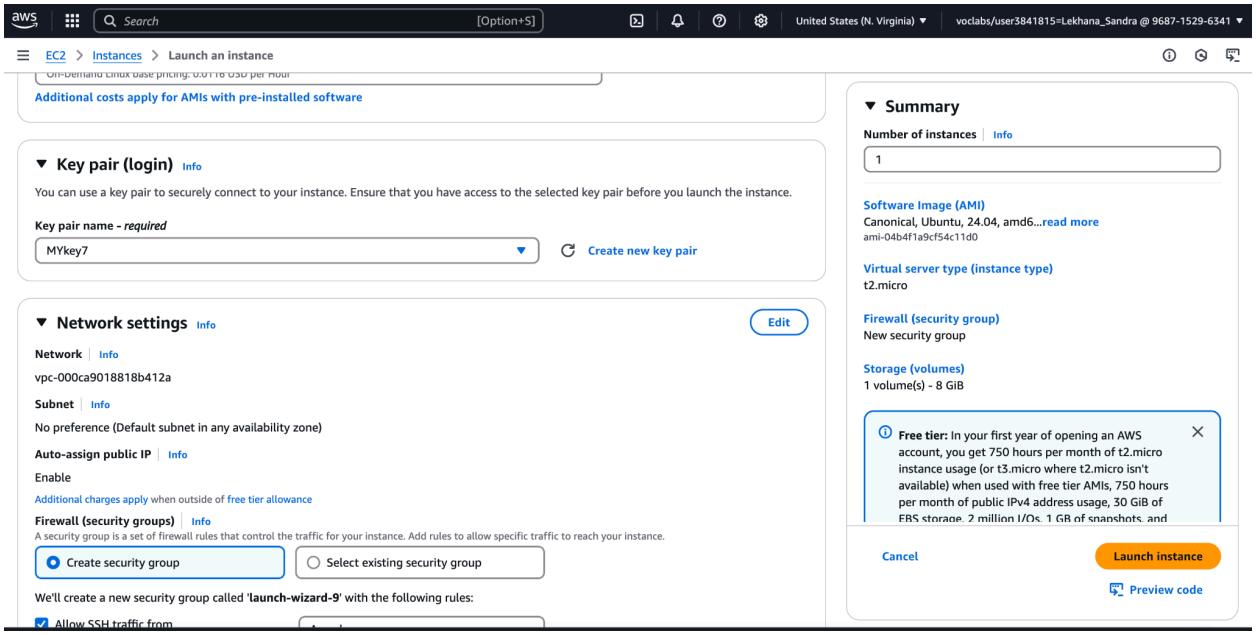
7. Copy the Provided Example and Execute in Terminal

Copy the given SSH connection command, for example:
ssh -i "MyKeyPair.pem" ubuntu@<Public_IP_of_Instance>

- Paste it into **Terminal** and press **Enter**.




Search
[Option+S]
United States (N. Virginia)
voclabs/user3841815=Lekhana_Sandra @ 9687-1529-6341
EC2 > Instances > Launch an instance
An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-04b4f1a9cf54c11d0 (64-bit (x86)) / ami-0a7a4e87939439934 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs
Free tier eligible
Description
Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).
Canonical, Ubuntu, 24.04, amd64 noble image
Architecture
64-bit (x86)
AMI ID
ami-04b4f1a9cf54c11d0
Username
ubuntu
Verified provider
Summary
Number of instances
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-04b4f1a9cf54c11d0
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of
Cancel
Launch instance
Preview code


Search
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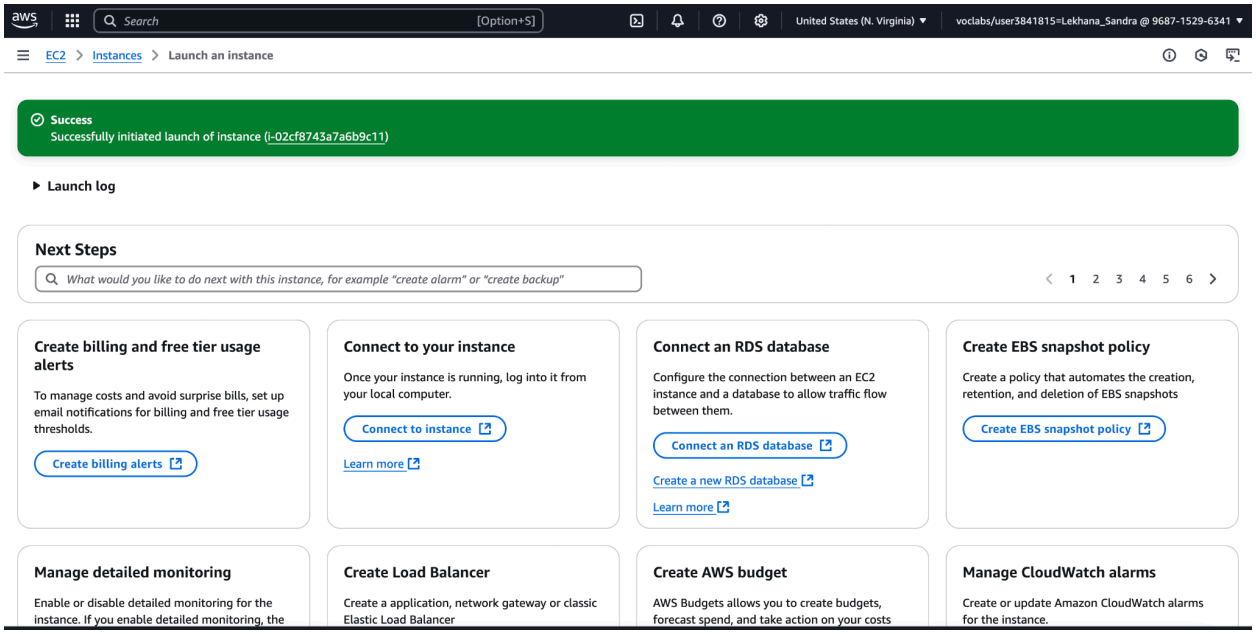
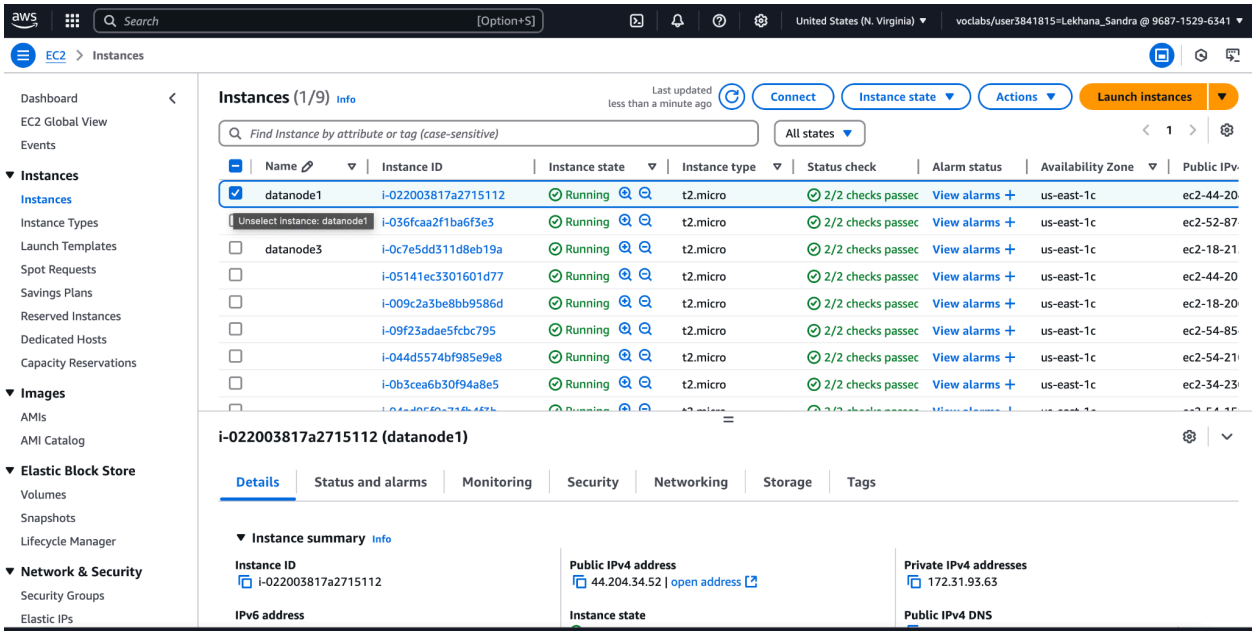
United States (N. Virginia)
voclabs/user3841815=Lekhana_Sandra @ 9687-1529-6341
EC2 > Instances > Launch an instance
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
MYkey7
Create new key pair
Network settings Info
Edit
Network Info
vpc-000ca9018818b412a
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Additional charges apply when outside of free tier allowance
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
We'll create a new security group called 'launch-wizard-9' with the following rules:
Allow SSH traffic from
Summary
Number of instances
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-04b4f1a9cf54c11d0
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of
Cancel
Launch instance
Preview code


Search
[Option+S]
United States (N. Virginia)
voclabs/user3841815=Lekhana_Sandra @ 9687-1529-6341
EC2 > Instances > Launch an instance
Success
Successfully initiated launch of instance (i-02cf8743a7a6b9c11)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
1 2 3 4 5 6
Create billing and free tier usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy
Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the
Create Load Balancer
Create a application, network gateway or classic Elastic Load Balancer
Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs
Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.


Search
[Option+S]
United States (N. Virginia)
voclabs/user3841815=Lekhana_Sandra @ 9687-1529-6341
EC2 > Instances
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Instances (1/9) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
datanode1 i-022003817a2715112 Running t2.micro 2/2 checks passed View alarms us-east-1c ec2-44-20
Unselect instance: datanode1
i-036fcaa2f1ba6f3e3 Running t2.micro 2/2 checks passed View alarms us-east-1c ec2-52-87
datanode3 i-0c7e5dd311d8eb19a Running t2.micro 2/2 checks passed View alarms us-east-1c ec2-18-21
i-05141ec3301601d77 Running t2.micro 2/2 checks passed View alarms us-east-1c ec2-44-20
i-009c2a3be8bb9586d Running t2.micro 2/2 checks passed View alarms us-east-1c ec2-18-20
i-09f23adae5fcbc795 Running t2.micro 2/2 checks passed View alarms us-east-1c ec2-54-85
i-044d5574bf985e9e8 Running t2.micro 2/2 checks passed View alarms us-east-1c ec2-54-21
i-0b3cea6b30f94a8e5 Running t2.micro 2/2 checks passed View alarms us-east-1c ec2-34-23
i-022003817a2715112 (datanode1)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-022003817a2715112
Public IPv4 address
44.204.34.52 | open address
Private IPv4 addresses
172.31.93.63
IPv6 address
Instance state
Public IPv4 DNS

aws | Search [Option+S] | United States (N. Virginia) | voclabs/user3841815=Lekhana_Sandra @ 9687-1529-6341

EC2 > Instances > i-022003817a2715112 > Connect to instance

Connect to instance Info

Connect to your instance i-022003817a2715112 (datanode1) using any of these options

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID
i-022003817a2715112 (datanode1)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is lab1.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
   chmod 400 "lab1.pem"
4. Connect to your instance using its Public DNS:
   ec2-44-204-34-52.compute-1.amazonaws.com

Example:
ssh -i "lab1.pem" ec2-user@ec2-44-204-34-52.compute-1.amazonaws.com

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

```
The authenticity of host 'ec2-3-95-229-238.compute-1.amazonaws.com (3.95.229.238)' can't be established.
ED25519 key fingerprint is SHA256:uPCvf/XdXVIusqikxMf4ZbNClNow0fdJSJRbOn7KdJA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-95-229-238.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Feb 21 21:34:20 UTC 2025

  System load:  0.0               Processes:             104
  Usage of /:   24.9% of 6.71GB   Users logged in:       0
  Memory usage: 20%               IPv4 address for enX0: 172.31.88.206
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```